## Summary of key points

**1** If the general term, $u_r$, of a series can be expressed in the form $f(r) - f(r+1)$

then $\sum_{r=1}^{n} u_r = \sum_{r=1}^{n} (f(r) - f(r+1))$.

so $u_1 = f(1) - f(2)$

$u_2 = f(2) - f(3)$

$u_3 = f(3) - f(4)$

$\vdots$

$u_n = f(n) - f(n+1)$

Then adding $\sum_{r=1}^{n} u_r = f(1) - f(n+1)$

**2** The **Maclaurin series** of a function $f(x)$ is given by

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(r)}(0)}{r!}x^r + \ldots$$

The series is valid provided that $f(0)$, $f'(0)$, $f''(0)$, … , $f^{(r)}(0)$, … all have finite values.

**3** The following Maclaurin series are given in the formulae booklet:

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots \qquad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots \qquad -1 < x \leqslant 1$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + (-1)^r \frac{x^{2r+1}}{(2r+1)!} + \ldots \qquad \text{for all } x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + (-1)^r \frac{x^{2r}}{(2r)!} + \ldots \qquad \text{for all } x$$

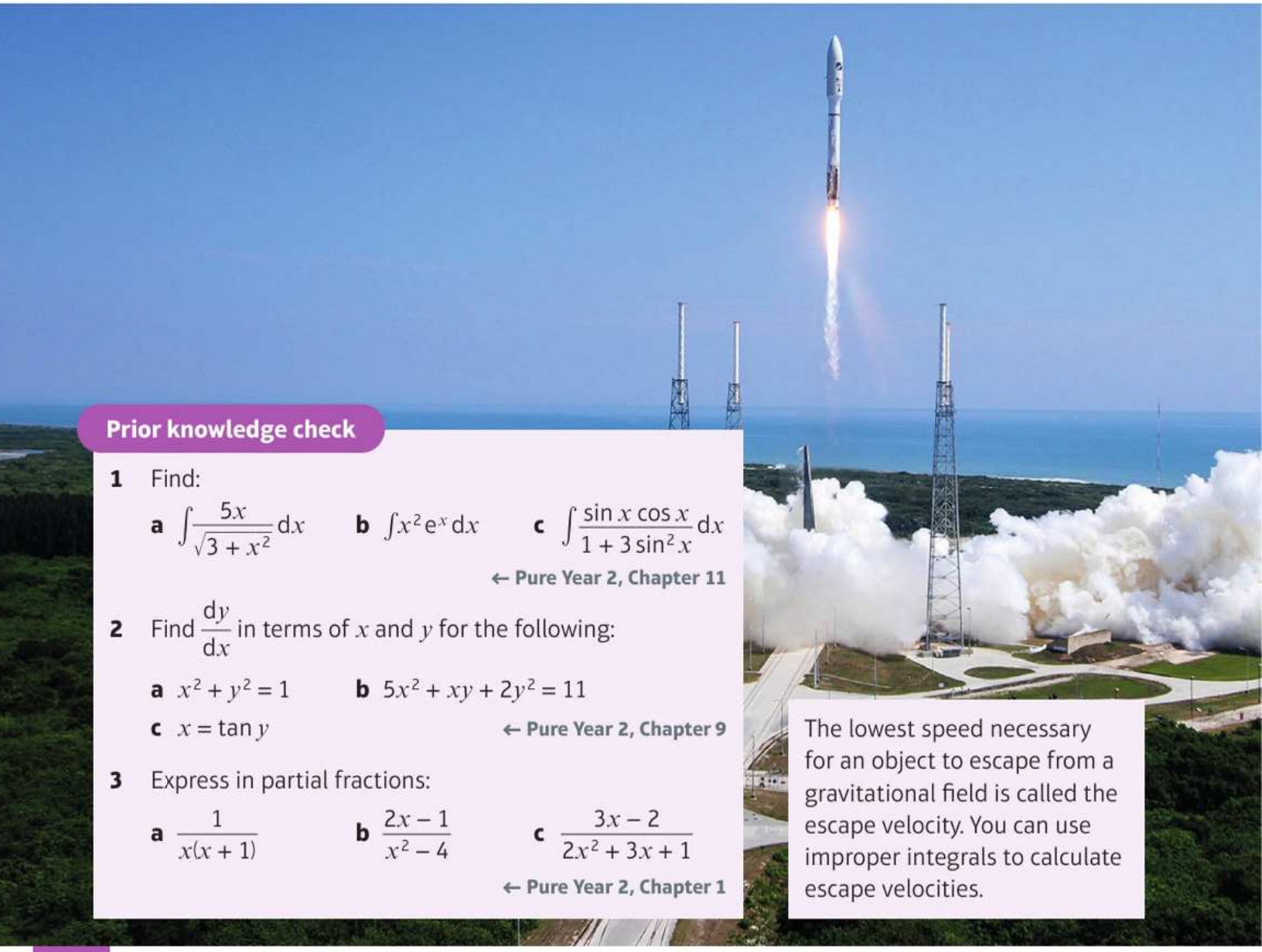
$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots + (-1)^r \frac{x^{2r+1}}{2r+1} + \ldots \qquad -1 \leqslant x \leqslant 1$$

# 3 Methods in calculus

## Objectives

After completing this chapter you should be able to:

- Evaluate improper integrals → pages 53–58
- Understand and evaluate the mean value of a function → pages 58–62
- Integrate rational functions using trigonometric substitutions → pages 62–69
- Integrate using partial fractions → pages 69–73



## Prior knowledge check

**1** Find:

**a** $\int \frac{5x}{\sqrt{3+x^2}}\,dx$ **b** $\int x^2 e^x\,dx$ **c** $\int \frac{\sin x \cos x}{1+3\sin^2 x}\,dx$

← Pure Year 2, Chapter 11

**2** Find $\frac{dy}{dx}$ in terms of $x$ and $y$ for the following:

**a** $x^2 + y^2 = 1$ **b** $5x^2 + xy + 2y^2 = 11$

**c** $x = \tan y$

← Pure Year 2, Chapter 9

**3** Express in partial fractions:

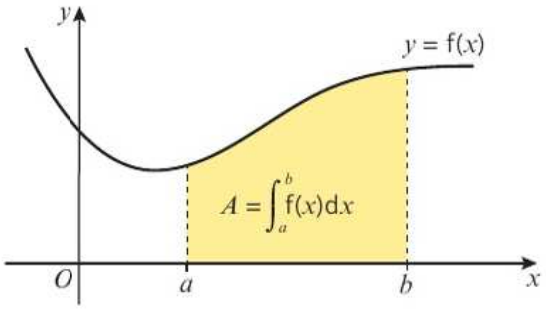
**a** $\frac{1}{x(x+1)}$ **b** $\frac{2x-1}{x^2-4}$ **c** $\frac{3x-2}{2x^2+3x+1}$

← Pure Year 2, Chapter 1

The lowest speed necessary for an object to escape from a gravitational field is called the escape velocity. You can use improper integrals to calculate escape velocities.

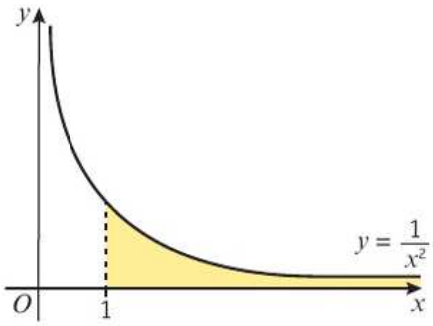
## 3.1 Improper integrals

If a function f($x$) exists and is continuous for all values in the interval $[a, b]$ then the definite integral $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$ represents the area enclosed by the curve $y = \mathrm{f}(x)$, the $x$-axis and the lines $x = a$ and $x = b$.

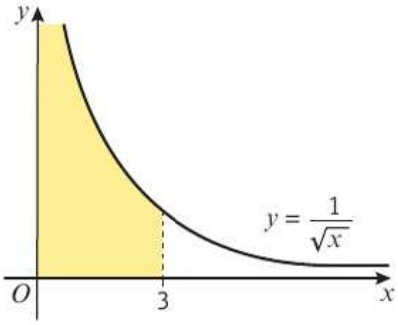



**Notation** The interval $[a, b]$ is all the real numbers $x$ satisfying the inequality $a \leqslant x \leqslant b$.

In this section you will consider integrals where one or both of the limits are **infinite**, or where the function is **not defined** at some point within the given interval. These are called **improper integrals**. In these cases it is still possible for the function to enclose a finite area.




The area bounded by the curve $y = \frac{1}{x^2}$, the $x$-axis and the line $x = 1$ is finite. This area is represented by the improper integral $\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x$.

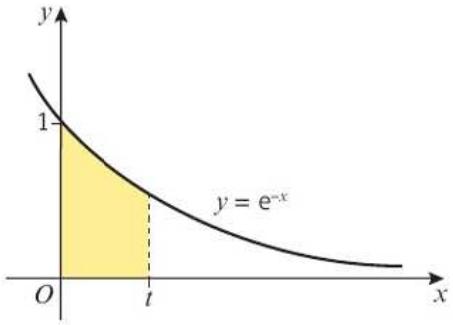


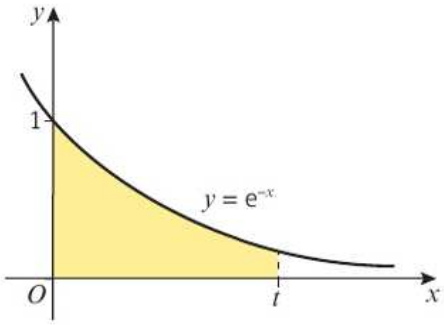

The function $\mathrm{f}(x) = \frac{1}{\sqrt{x}}$ is not defined at $x = 0$. However, the area bounded by the curve $y = \frac{1}{\sqrt{x}}$, the coordinate axes and the line $x = 3$ is finite. This area is represented by the improper integral $\int_0^3 \frac{1}{\sqrt{x}}\,\mathrm{d}x$.

- **The integral $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$ is improper if either:**
  - **one or both of the limits is infinite**
  - **f($x$) is undefined at $x = a$, $x = b$ or another point in the interval $[a, b]$.**

**Notation** If an improper integral exists then it is said to be **convergent**. If it does not exist it is said to be **divergent**.

You can determine whether improper integrals are convergent, and evaluate them if so, by considering limits. To find $I = \int_0^\infty \mathrm{e}^{-x}\,\mathrm{d}x$, you need to consider the integral $\int_0^t \mathrm{e}^{-x}\,\mathrm{d}x$, for some finite value $t$. If this integral tends to a limit as $t \to \infty$ then $I$ is convergent and equal to that limit. If it fails to tend to a limit, then $I$ is divergent.




As $t \to \infty$,

$\int_0^t e^{-x}\,dx \to \int_0^\infty e^{-x}\,dx$

$\int_0^t e^{-x}\,dx = [-e^{-x}]_0^t = -e^{-t} + 1$

Since $e^{-t} \to 0$ as $t \to \infty$, the integral $\int_0^t e^{-x}\,dx \to 1$ as $t \to \infty$.

So $\int_0^\infty e^{-x}\,dx$ is convergent and equal to 1.

**Notation** You can use limit notation to write:

$$\int_0^\infty e^{-x}\,dx = \lim_{t\to\infty}\int_0^t e^{-x}\,dx = \lim_{t\to\infty}(-e^{-t} + 1) = 1$$

## Example 1

Evaluate each improper integral, or show that it is not convergent.

**a** $\int_1^\infty \frac{1}{x^2}\,dx$ **b** $\int_1^\infty \frac{1}{x}\,dx$

**a** $\int_1^\infty \frac{1}{x^2}\,dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^2}\,dx$

$= \lim_{t\to\infty}\left[-\frac{1}{x}\right]_1^t$

$= \lim_{t\to\infty}\left(-\frac{1}{t} + 1\right)$

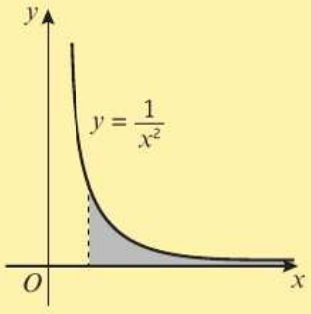
$= 1$

So $\int_1^\infty \frac{1}{x^2}\,dx$ converges and $\int_1^\infty \frac{1}{x^2}\,dx = 1$

Replace the infinite limit by $t$, and take the limit as $t \to \infty$.

$\frac{1}{t} \to 0$ as $t \to \infty$

The area under the curve $y = \frac{1}{x^2}$ from 1 to infinity is finite and is exactly equal to 1.




**b** $\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\int_1^t \frac{1}{x}\,dx$

$= \lim_{t\to\infty}[\ln x]_1^t$

$= \lim_{t\to\infty}(\ln t - \ln 1)$

$= \lim_{t\to\infty}(\ln t)$

$\ln t \to \infty$ as $t \to \infty$, so $\int_1^\infty \frac{1}{x}\,dx$ does not converge.

**Online** Explore the integral $\int_1^\infty \frac{1}{x^2}\,dx$ using GeoGebra.



**Watch out** Make sure you show the limiting process clearly in your working. You can't just write $\int_1^\infty \frac{1}{x}\,dx = \ln\infty = \infty$.

You need to use a similar limiting process if the function you are integrating is not defined at one or more points in the interval.

## Example 2

Evaluate each integral, or show that it does not converge.

**a** $\int_0^1 \frac{1}{x^2}\,dx$ **b** $\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx$

**a** $\int_0^1 \frac{1}{x^2}\,dx = \lim_{t\to 0}\int_t^1 \frac{1}{x^2}\,dx$

$= \lim_{t\to 0}\left[-\frac{1}{x}\right]_t^1$

$= \lim_{t\to 0}\left(-1+\frac{1}{t}\right)$

$-1+\frac{1}{t}\to\infty$ as $t\to 0$, so $\int_0^1 \frac{1}{x^2}\,dx$ does not converge.

$y = \frac{1}{x^2}$ is undefined for $x = 0$, so replace the lower limit with $t$ and take the limit as $t \to 0$.

$\frac{1}{t}\to\infty$ as $t\to 0$

**b** $\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx = \lim_{t\to 2}\int_0^t \frac{x}{\sqrt{4-x^2}}\,dx$

$= \lim_{t\to 2}\left[-\sqrt{4-x^2}\right]_0^t$

$= \lim_{t\to 2}\left(-\sqrt{4-t^2}-(-\sqrt{4})\right)$

$= \lim_{t\to 2}\left(2-\sqrt{4-t^2}\right)$

$= 2$

So $\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx$ converges and

$\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx = 2$

$y = \frac{x}{\sqrt{4-x^2}}$ is not defined for $x = 2$, so replace the upper limit with $t$.

$\lim_{t\to 2}\left(\sqrt{4-t^2}\right) = 0$

If **both limits** of an integral are infinite, then you need to split the integral into the sum of two improper integrals. In other words, you write $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx$ for some value $c$. If both of these integrals converge, then the original integral converges, but if either diverges, then the original integral is also divergent.

**Watch out** Do not write $\int_{-\infty}^{\infty} f(x)\,dx$ as $\lim_{a\to\infty}\int_{-a}^{a} f(x)\,dx$. You must split it into two separate integrals to determine whether it converges.

→ Review Exercise 1, Challenge Q3

## Example 3

**a** Find $\int x e^{-x^2}\,dx$. **b** Hence show that $\int_{-\infty}^{\infty} x e^{-x^2}\,dx$ converges and find its value.

**a** Let $I = \int x e^{-x^2}\,dx$

Consider $y = e^{-x^2}$

$\frac{dy}{dx} = -2x e^{-x^2}$

So $I = -\frac{1}{2}e^{-x^2} + c$

Try differentiating $e^{-x^2}$.

The $-2x$ comes from differentiating $-x^2$.

Use the reverse chain rule.

**b** Split $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$ up as $\int_{-\infty}^{0} xe^{-x^2}\,dx + \int_{0}^{\infty} xe^{-x^2}\,dx$

Consider $\int_{-t}^{0} xe^{-x^2}\,dx$:

$\int_{-t}^{0} xe^{-x^2}\,dx = \left[-\frac{1}{2}e^{-x^2}\right]_{-t}^{0} = -\frac{1}{2} + \frac{1}{2}e^{-t^2}$

So $\lim_{t\to\infty} \int_{-t}^{0} xe^{-x^2}\,dx = \lim_{t\to\infty}\left(-\frac{1}{2} + \frac{1}{2}e^{-t^2}\right) = -\frac{1}{2}$

So $\int_{-\infty}^{0} xe^{-x^2}\,dx$ converges and $\int_{-\infty}^{0} xe^{-x^2}\,dx = -\frac{1}{2}$

Similarly, consider $\int_{0}^{t} xe^{-x^2}\,dx$:

$\int_{0}^{t} xe^{-x^2}\,dx = \left[-\frac{1}{2}e^{-t^2}\right]_{0}^{t} = -\frac{1}{2}e^{-t^2} + \frac{1}{2}$

So $\lim_{t\to\infty} \int_{0}^{t} xe^{-x^2}\,dx = \lim_{t\to\infty}\left(-\frac{1}{2}e^{-t^2} + \frac{1}{2}\right) = \frac{1}{2}$

So $\int_{0}^{\infty} xe^{-x^2}\,dx$ converges and $\int_{0}^{\infty} xe^{-x^2}\,dx = \frac{1}{2}$

Since both integrals converge, we know that $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$ converges and

$\int_{-\infty}^{\infty} xe^{-x^2}\,dx = \int_{-\infty}^{0} xe^{-x^2}\,dx + \int_{0}^{\infty} xe^{-x^2}\,dx$

$= -\frac{1}{2} + \frac{1}{2} = 0$

**Problem-solving**

You can choose any point at which to split the integral up, but choosing a special value like 0 will often make evaluating the integral easier.

To find the integral between $-\infty$ and 0, you should find the integral between $-t$ and 0 and then let $t \to \infty$.

Use $\int xe^{-x^2}\,dx = -\frac{1}{2}e^{-x^2} + c$

**Watch out** You need to check that **both** the integrals converge before you can determine that the original integral converges.

To find the integral between 0 and $\infty$, you should find the integral between 0 and $t$ and then let $t \to \infty$.

Use $\int xe^{-x^2}\,dx = -\frac{1}{2}e^{-x^2} + c$

Apply $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_{c}^{\infty} f(x)\,dx$

## Exercise 3A

**1** Find the values of the following improper integrals.

**a** $\int_{1}^{\infty} \frac{1}{x^3}\,dx$ **b** $\int_{2}^{\infty} x^{-\frac{3}{2}}\,dx$ **c** $\int_{0}^{\infty} e^{-3x}\,dx$

**2** For each of the following, show that the improper integral diverges.

**a** $\int_{0}^{\infty} e^{x}\,dx$ **b** $\int_{1}^{\infty} \frac{1}{\sqrt{x}}\,dx$ **c** $\int_{0}^{\infty} \frac{8x}{\sqrt{1+x^2}}\,dx$

**3** For each of the following, show that the improper integral converges and find its value.

**a** $\int_{0}^{1} \frac{1}{\sqrt{x}}\,dx$ **b** $\int_{0}^{\frac{2}{3}} \frac{1}{\sqrt{2-3x}}\,dx$ **c** $\int_{0}^{\ln 3} \frac{e^{x}}{\sqrt{e^{x}-1}}\,dx$

(P) **4** For each of the following, determine whether the integral converges, and if so, find its value.

**a** $\int_{-1}^{2} \frac{1}{\sqrt{|x|}}\,dx$ **b** $\int_{-1}^{3} \frac{x-1}{\sqrt{3+2x-x^2}}\,dx$ **c** $\int_{0}^{\pi} \tan x\,dx$

E **5** **a** Find $\int \frac{1}{(7-3x)^2}\,dx$. **(2 marks)**

**b** Hence show that $\int_{-\infty}^{2} \frac{1}{(7-3x)^2}\,dx$ converges and find its value. **(3 marks)**

**Watch out** Make sure you show the limiting process clearly in your working.

E **6** **a** Find $\int x^2 e^{x^3}\,dx$. **(2 marks)**

**b** Hence show that $\int_{-\infty}^{1} x^2 e^{x^3}\,dx$ converges and find its value. **(3 marks)**

E **7** **a** Find $\int \frac{\ln x}{x}\,dx$. **(3 marks)**

**b** Hence show that $\int_{1}^{\infty} \frac{\ln x}{x}\,dx$ is divergent. **(3 marks)**

E **8** **a** Find $\int (\ln x)^2\,dx$. **(2 marks)**

Hence show that:

**b** $\int_{0}^{1} (\ln x)^2\,dx$ is convergent. **(2 marks)**

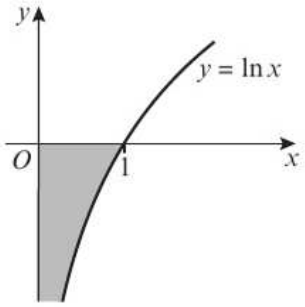
**c** $\int_{0}^{\infty} (\ln x)^2\,dx$ is divergent. **(2 marks)**

E **9** Evaluate $\int_{0}^{2} \frac{6x}{\sqrt[3]{4-x^2}}\,dx$. **(4 marks)**

E/P **10** Evaluate $\int_{-2}^{2} \frac{\sqrt{2-x} - 3\sqrt{2+x}}{\sqrt{4-x^2}}\,dx$. **(5 marks)**

E/P **11** The diagram shows the curve with equation $y = \ln x$.

Find the shaded area enclosed by the curve and the coordinate axes. You may assume that $x \ln x \to 0$ as $x \to 0$. **(5 marks)**




E **12** **a** Explain why $\int_{0}^{\frac{\pi}{2}} \tan x\,dx$ is an improper integral. **(1 mark)**

**b** Show that $\int_{0}^{\frac{\pi}{2}} \tan x\,dx$ is divergent. **(3 marks)**

$$\int_{x=\ln 2}^{x=\ln 6} \frac{4}{1+e^x}\,dx = \int_{u=2}^{u=6} \left(\frac{4}{1+u}\right)\left(\frac{du}{u}\right)$$

$$= 4\int_2^6 \frac{1}{(1+u)u}\,du$$

Transform the integral by substituting $u = e^x$ and $\frac{du}{dx} = e^x$ and substituting the limits.

Using partial fractions:

$$\frac{1}{u(1+u)} = \frac{A}{u} + \frac{B}{1+u}$$

To calculate $4\int_2^6 \frac{1}{(1+u)u}\,du$ it is necessary to use partial fractions.

So $A(1+u) + Bu = 1$

When $u = 0$, $A = 1$.

Equating $u$ terms, $A + B = 0$, so $B = -1$.

Find the values of $A$ and $B$.

$$4\int_2^6 \frac{1}{(1+u)u}\,du = 4\int_2^6 \left(\frac{1}{u} - \frac{1}{1+u}\right)du$$

$$= 4[\ln u - \ln(u+1)]_2^6$$

Integrate with respect to $u$.

$$= 4((\ln 6 - \ln 7) - (\ln 2 - \ln 3))$$

Evaluate the integral using the limits.

$$= 4\ln\frac{18}{14}$$

$$= 4\ln\frac{9}{7}$$

Simplify using the laws of logarithms.

So the mean value of f on $[\ln 2, \ln 6]$ is

$$\frac{1}{\ln 6 - \ln 2}\left(4\ln\frac{9}{7}\right) = \frac{4\ln\frac{9}{7}}{\ln 3}$$

Find $\frac{1}{b-a}\int_a^b f(x)\,dx$ and use the laws of logarithms to simplify.

**b** $$\int_{\ln 2}^{\ln 6}\left(\frac{4}{1+e^x} + 4\right)dx = \int_{\ln 2}^{\ln 6}\frac{4}{1+e^x}\,dx + \int_{\ln 2}^{\ln 6} 4\,dx$$

The original integral can be separated.

$$\int_{\ln 2}^{\ln 6} 4\,dx = [4x]_{\ln 2}^{\ln 6}$$

$$= 4(\ln 6 - \ln 2)$$

$$= 4\ln 3$$

Calculate the new integral.

Calculating the mean value,

$$\frac{1}{\ln 3}\int_{\ln 2}^{\ln 6}\left(\frac{4}{1+e^x} + 4\right)dx$$

$$= \frac{1}{\ln 3}\int_{\ln 2}^{\ln 6}\frac{4}{1+e^x}\,dx + \frac{1}{\ln 3}\int_{\ln 2}^{\ln 6} 4\,dx$$

$$= \frac{4\ln\frac{9}{7}}{\ln 3} + \frac{1}{\ln 3}(4\ln 3)$$

$$= \frac{4\ln\frac{9}{7}}{\ln 3} + 4$$

**c** $-f(x)$ is a reflection in the $x$-axis of $f(x)$.

The mean value of $f(x)$ over the interval $[\ln 2, \ln 6]$ was $\frac{4\ln\frac{9}{7}}{\ln 3}$

Therefore, the mean value of $-f(x)$ over the interval $[\ln 2, \ln 6]$ is $-\frac{4\ln\frac{9}{7}}{\ln 3}$

**Problem-solving**

Every value of $f(x)$ in the interval has increased by 4, so the mean value has increased by 4.

In the example above, you saw that geometric considerations can be used to find mean values of transformed functions, if you already know the mean value of the original function on the same interval.

**If the function f($x$) has mean value $\bar{f}$ over the interval $[a, b]$, and $k$ is a real constant, then:**

- **f($x$) + $k$ has mean value $\bar{f} + k$ over the interval $[a, b]$**
- **$k$f($x$) has mean value $k\bar{f}$ over the interval $[a, b]$**
- **−f($x$) has mean value $-\bar{f}$ over the interval $[a, b]$.**

**Watch out** You cannot deduce the mean value of f(−$x$) or f($kx$) in this way.

## Exercise 3B

**1** For each of the following functions f($x$), find the mean value of f($x$) on [0, 1].

**a** f($x$) = 1 **b** $f(x) = \frac{1}{x+1}$ **c** $f(x) = e^x + 1$

**2** Find the exact mean value of f($x$) over the given interval.

**a** $f(x) = \frac{e^{3x}}{e^{3x}+1}$; [0, 2] **b** $f(x) = \cos^3 x \sin^2 x$; $\left[0, \frac{\pi}{2}\right]$

**c** $f(x) = xe^{-x}$; [1, 3] **d** $f(x) = \frac{5}{(x+2)(2x+1)}$; [0, 3]

**e** $f(x) = (\sec x - \cos x)^2$; $\left[0, \frac{\pi}{4}\right]$.

**3** $f(x) = x^3 - 3x^2 - 24x + 100$

**a** Find the coordinates of the turning points of f($x$).

**b** Sketch the graph of $y$ = f($x$).

**c** Without calculation, state an upper and lower bound on the mean value of the function on the interval [−2, 4], giving a reason for your answer.

**d** Calculate the exact mean value of f($x$) over the interval [−2, 4].

(E) **4** Find the exact mean value of $f(x) = \frac{\sin x \cos x}{\cos 2x + 2}$ over the interval $\left[0, \frac{\pi}{2}\right]$. **(4 marks)**

(E) **5** Find the exact mean value of $f(x) = x\sqrt{x+4}$ over the interval [0, 5]. **(4 marks)**

(E) **6** Find the exact mean value of $f(x) = x \sin 2x$ over the interval $\left[0, \frac{\pi}{3}\right]$. **(4 marks)**

(E/P) **7** $f(x) = \frac{5x}{(2x-1)(x+2)}$

**a** Show that the mean value of f($x$) over the interval [1, 5] is $\frac{1}{4}\ln\frac{49}{3}$ **(4 marks)**

**b** Hence, or otherwise, find the mean value over the interval [1, 5] of f($x$) + ln $k$ where $k$ is a positive constant, giving your answer in the form $p \ln q$, where $p$ and $q$ are constants and $q$ is in terms of $k$. **(2 marks)**

(E/P) **8** $f(x) = x(x^2 - 4)^4$

**a** Show that the mean value of f($x$) over the interval [0, 2] is $\frac{256}{5}$ **(3 marks)**

**b** Use the answer to part **a** to find the mean value over the interval [0, 2] of −2f($x$). **(2 marks)**

(E/P) **9** f($x$) = ln($kx$), where $k$ is a positive constant.

Given that the mean value of f($x$) on the interval [0, 2] is −2, find the value of $k$. **(4 marks)**

(P) **10** Prove that if $f(x)$ has mean value $m$ on the interval $[a, b]$, then $f(x) + c$ has mean value $m + c$.

(E/P) **11** $f(x) = \dfrac{1}{\sqrt{2 - x}}$

**Watch out** This is an improper integral.

Find the exact mean value of $f(x)$ on the interval $[0, 2]$. **(6 marks)**

(P) **12** Use geometric reasoning to explain why the mean value of $f(x) = \sin^5 x$ on the interval $[0, 2\pi]$ is 0.

(E/P) **13** $f(x) = \dfrac{\cos x}{(2 + \sin x)^2}$

**a** Find $\int f(x)\,dx$. **(4 marks)**

**b** Hence show that the mean value of $f(x)$ over the interval $\left[0, \frac{5\pi}{3}\right]$ is $-\dfrac{3}{130\pi}(3 + 4\sqrt{3})$. **(2 marks)**

**c** Hence, or otherwise, find the mean value, over the interval $\left[0, \frac{5\pi}{3}\right]$, of $f(x) + 3x$. **(3 marks)**

(P) **14** **a** Sketch a graph of $f(x) = 1 - 3x - 2x^2$, finding the coordinates of any turning points.

**b** Calculate $\int_a^{a+1} f(x)\,dx$ for $a \in \mathbb{R}$.

**c** Find the maximum possible mean value of $f(x)$ on any real interval of length 1.

## 3.3 Differentiating inverse trigonometric functions

You can differentiate the inverse trigonometric functions implicitly.

### Example 6

Show that $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1 - x^2}}$

Let $y = \arcsin x$

then $\sin y = x$

$\cos y \dfrac{dy}{dx} = 1$

$\dfrac{dy}{dx} = \dfrac{1}{\cos y}$

$= \dfrac{1}{\sqrt{1 - \sin^2 y}}$

but $\sin y = x$

so $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 - x^2}}$

Use $y = \sin(\arcsin x) = x$.

Differentiate implicitly. You could also differentiate $x = \sin y$ with respect to $y$ then use $\dfrac{dy}{dx} = \dfrac{1}{\frac{dx}{dy}}$

← Pure Year 2, Section 9.6

Divide by $\cos y$.

Use $\cos^2 y = 1 - \sin^2 y$ and that $\cos y \geqslant 0$ when $y$ is in the range of arcsin, i.e. $-\dfrac{\pi}{2} \leqslant y \leqslant \dfrac{\pi}{2}$

**Problem-solving**

Alternatively, since $\dfrac{dy}{dx} = \pm\dfrac{1}{\sqrt{1 - x^2}}$ you can conclude the sign is positive since the graph of $y = \arcsin x$ shows that the gradient is positive at all points $x$.

You can use similar methods to obtain the following standard results.

- $\mathbf{\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}}$
- $\mathbf{\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}}$
- $\mathbf{\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}}$

You should learn these results, but also be able to derive them as in the example above.

Notice that $\frac{d}{dx}(\arcsin x) = -\frac{d}{dx}(\arccos x)$.

**Links** Recall the domain and range of each of these inverse trigonometric functions:

| Function | Domain | Range |
|---|---|---|
| $\arcsin x$ | $[-1, 1]$ | $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ |
| $\arccos x$ | $[-1, 1]$ | $[0, \pi]$ |
| $\arctan x$ | $(-\infty, \infty)$ | $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ |

← Pure Year 2, Chapter 6

## Example 7

Given $y = \arcsin x^2$, find $\frac{dy}{dx}$

**a** using implicit differentiation

**b** using the chain rule and the formula for $\frac{d}{dx}\arcsin x$.

**a** $\sin y = x^2$ — Use $\sin(\arcsin x) = x$.

$\cos y \frac{dy}{dx} = 2x$ — Differentiate.

$\frac{dy}{dx} = \frac{2x}{\cos y}$ — Divide by $\cos y$.

$\frac{dy}{dx} = \frac{2x}{\sqrt{1-\sin^2 y}}$ — Use $\cos^2 y = 1 - \sin^2 y$ and that $\cos y \geqslant 0$ when $y$ is in the range of arcsin, i.e. $-\frac{\pi}{2} \leqslant y \leqslant \frac{\pi}{2}$

but $\sin y = x^2$

so $\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^4}}$

**b** Let $t = x^2$, then $y = \arcsin t$ — Substitute $t = x^2$ to get $\arcsin x^2$ in the form $\arcsin x$.

Then $\frac{dt}{dx} = 2x$ and $\frac{dy}{dt} = \frac{1}{\sqrt{1-t^2}}$ — Differentiate $t$ and $y$.

$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ — Use the chain rule.

$= \frac{2x}{\sqrt{1-x^4}}$

## Example 8

Given $y = \arctan\left(\frac{1-x}{1+x}\right)$, find $\frac{dy}{dx}$

$\tan y = \frac{1-x}{1+x}$ — Use $\tan(\arctan x) = x$.

$\sec^2 y \frac{dy}{dx} = \frac{-(1+x) - 1(1-x)}{(1+x)^2}$ — Differentiate using the quotient rule on $\frac{1-x}{1+x}$

$= -\frac{2}{(1+x)^2}$ — Simplify.

So $\frac{dy}{dx} = \frac{1}{\sec^2 y} \times \left(-\frac{2}{(1+x)^2}\right)$ — Divide by $\sec^2 y$.

$= \frac{1}{1+\tan^2 y} \times \left(-\frac{2}{(1+x)^2}\right)$ — Use $1 + \tan^2 y \equiv \sec^2 y$.

$= \frac{1}{\left(1+\frac{1-x}{1+x}\right)^2} \times \left(-\frac{2}{(1+x)^2}\right)$ — Substitute $\tan y = \frac{1-x}{1+x}$

$= \frac{(1+x)^2}{(1+x)^2 + (1-x)^2} \times \left(-\frac{2}{(1+x)^2}\right)$ — Cancel $(1+x)^2$.

$= -\frac{2}{2+2x^2}$

$= -\frac{1}{1+x^2}$

**Problem-solving**

You could also use the chain rule and the formula for $\frac{d}{dx}(\arctan x)$.

## Exercise 3C

**1** Use implicit differentiation to differentiate the following functions.

**a** $\arctan x$ **b** $\arccos x$

**c** $\arccos x^2$ **d** $\arctan(x^3 + 3x)$

**e** $\arcsin\left(\frac{1}{x}\right)$

**Hint** You can check your answers by using the chain rule as well as the results stated earlier in the section.

**2** Differentiate $y = (\arccos x)(\arcsin x)$

(E) **3** Differentiate $y = \frac{1 + \arctan x}{1 - \arctan x}$ **(4 marks)**

(E/P) **4** $f(x) = \arccos x + \arcsin x$

By considering $\int f'(x)\,dx$, prove that $f(x) = \frac{\pi}{2}$ for all values of $x$. **(4 marks)**

**5** Differentiate with respect to $x$:

**a** $\arccos 2x$ **b** $\arctan \frac{x}{2}$ **c** $\arcsin 3x$ **d** $\operatorname{arcot}(x+1)$

**e** $\arcsin(1 - x^2)$ **f** $\arccos x^2$ **g** $e^x \arccos x$ **h** $\arcsin x \cos x$

**i** $x^2 \arccos x$ **j** $e^{\arctan x}$

E **6** Given that $\tan y = x \arctan x$, find $\dfrac{dy}{dx}$ **(4 marks)**

E **7** Given that $y = \arcsin x$, prove that

$$(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$$ **(6 marks)**

**8** Find an equation of the tangent to the curve with equation $y = \arcsin 2x$ at the point where $x = \frac{1}{4}$

**9** Find the derivatives of the following functions.

**a** $(\arctan x)^2$ **b** $\dfrac{1}{\arcsin x}$ **c** $\arctan(\arctan x)$

P **10** Sketch the graphs of the following:

**a** $\arcsin(\arcsin x)$ **b** $\arccos(\arccos x)$ **c** $\arctan(\arctan x)$

P **11** Prove each of the following:

**a** $\sin(\arccos x) = \sqrt{1 - x^2}$ **b** $\cos(\arctan x) = \dfrac{1}{\sqrt{1 + x^2}}$

**c** $\sec(\arccos x) = \dfrac{1}{x}$ **d** $\sin(\operatorname{arcsec} x) = \sqrt{1 - \dfrac{1}{x^2}}$

**Hint** When taking a square root, you should consider the definitions or graphs of the inverse trigonometric functions to determine the sign.

## 3.4 Integrating with inverse trigonometric functions

You can use the results from the previous section to integrate functions of the forms $\dfrac{1}{a^2 + x^2}$ and $\dfrac{1}{\sqrt{a^2 - x^2}}$

### Example 9

By using an appropriate substitution, show that $\displaystyle\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + c$, where $a$ is a positive constant and $|x| < a$.

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \int \frac{1}{\sqrt{a^2\left(1 - \left(\frac{x}{a}\right)^2\right)}}\,dx$$

$$= \frac{1}{a}\int \frac{1}{\sqrt{1 - \left(\frac{x}{a}\right)^2}}\,dx$$

$$= \int \frac{1}{\sqrt{1 - u^2}}\,du$$

Use the substitution $u = \dfrac{x}{a}$ and $du = \dfrac{dx}{a}$

$$= \arcsin u + c$$

Recall that $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1 - x^2}}$

$$= \arcsin\left(\frac{x}{a}\right) + c$$

The following two results are given in the formula booklet. You can quote them when calculating other integrals, but you should also be able to derive them using substitution as in the example above.

- $\int \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c,\ a > 0,\ |x| < a$
- $\int \frac{1}{\sqrt{a^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{a}\right) + c$

## Example 10

Find $\int \frac{4}{5 + x^2}\,\mathrm{d}x$.

$$\int \frac{4}{5 + x^2}\,dx = 4\int \frac{1}{5 + x^2}\,dx$$

$$= 4\left(\frac{1}{\sqrt{5}}\arctan\left(\frac{x}{\sqrt{5}}\right)\right) + c$$

$$= \frac{4}{\sqrt{5}}\arctan\left(\frac{x}{\sqrt{5}}\right) + c$$

Use $\int \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$ with $a = \sqrt{5}$.

## Example 11

**a** Find $\int \frac{1}{25 + 9x^2}\,\mathrm{d}x$.

**b** Evaluate $\int_{-\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{4}} \frac{1}{\sqrt{3 - 4x^2}}$, leaving your answer in terms of $\pi$.

**a**

$$\int \frac{1}{25 + 9x^2}\,dx = \int \frac{1}{9\left(\frac{25}{9} + x^2\right)}\,dx$$

$$= \frac{1}{9}\left(\left(\frac{1}{\left(\frac{5}{3}\right)}\right)\arctan\left(\frac{x}{\left(\frac{5}{3}\right)}\right)\right) + c$$

$$= \frac{1}{15}\arctan\left(\frac{3x}{5}\right) + c$$

You need to write $25 + 9x^2$ in the form $k(a^2 + x^2)$.

Use $\int \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$ with $a = \frac{5}{3}$

**b**

$$\int_{-\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{4}} \frac{1}{\sqrt{3 - 4x^2}}\,dx = \int_{-\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{4}} \frac{1}{\sqrt{4\left(\frac{3}{4} - x^2\right)}}\,dx$$

$$= \frac{1}{2}\int_{-\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{4}} \frac{1}{\sqrt{\left(\frac{3}{4} - x^2\right)}}\,dx$$

$$= \frac{1}{2}\left[\arcsin\left(\frac{2x}{\sqrt{3}}\right)\right]_{-\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{4}}$$

$$= \frac{1}{2}\arcsin\left(\frac{1}{2}\right) - \frac{1}{2}\arcsin\left(-\frac{1}{2}\right)$$

$$= \frac{\pi}{12} - \left(-\frac{\pi}{12}\right)$$

$$= \frac{\pi}{6}$$

Write $3 - 4x^2$ in the form $k(a^2 - x^2)$.

Use $\int \frac{1}{\sqrt{a^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{a}\right) + c$ with $a = \frac{\sqrt{3}}{2}$

$-\frac{\pi}{2} \leqslant \arcsin x \leqslant \frac{\pi}{2}$

so $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$ and $\arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$

## Example 12

Find $\int \frac{x+4}{\sqrt{1-4x^2}}\,dx$.

$$\int \frac{x+4}{\sqrt{1-4x^2}}\,dx = \int \frac{x}{\sqrt{1-4x^2}}\,dx + 4\int \frac{1}{\sqrt{1-4x^2}}\,dx$$

To calculate $\int \frac{x}{\sqrt{1-4x^2}}\,dx$ use the substitution

$u = 1 - 4x^2$

$du = -8x\,dx$

$$\int \frac{x}{\sqrt{1-4x^2}}\,dx = -\tfrac{1}{8}\int \frac{1}{\sqrt{u}}\,du$$

$$= -\tfrac{1}{8}\int u^{-\frac{1}{2}}\,du$$

$$= -\tfrac{1}{4}u^{\frac{1}{2}} + c$$

$$= -\tfrac{1}{4}\sqrt{1-4x^2} + c$$

$$4\int \frac{1}{\sqrt{1-4x^2}}\,dx = 4\int \frac{1}{\sqrt{4\left(\frac{1}{4}-x^2\right)}}\,dx$$

$$= 2\int \frac{1}{\sqrt{\frac{1}{4}-x^2}}\,dx$$

$$= 2\arcsin 2x + c$$

So $\int \frac{x+4}{\sqrt{1-4x^2}}\,dx = 2\arcsin 2x - \tfrac{1}{4}\sqrt{1-4x^2} + c$

**Problem-solving**

The integral can be split into a fraction which can be integrated using the 'reverse chain rule' and one that looks like those covered above.

Substitute for $x$ and $dx$ and adjust for the constant.

Rewrite in terms of $x$ by resubstituting $u = 1 - 4x^2$.

Write $1 - 4x^2$ in the form $k(a^2 - x^2)$.

Use $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + c$ with $a = \frac{1}{2}$

## Exercise 3D

**1** Use the substitution $x = a\tan\theta$ to show that $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$.

**2** Use the substitution $x = \cos\theta$ to show that $\int \frac{1}{\sqrt{1-x^2}}\,dx = -\arccos x + c$.

**3** Find:

**a** $\int \frac{3}{\sqrt{4-x^2}}\,dx$ **b** $\int \frac{4}{5+x^2}\,dx$ **c** $\int \frac{1}{\sqrt{25-x^2}}\,dx$ **d** $\int \frac{1}{\sqrt{x^2-2}}\,dx$

(E) **4** Find $\int \frac{1}{4+3x^2}\,dx$, giving your answer in the form $A\arctan(Bx) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(3 marks)**

E **5** Show that $\int \frac{1}{\sqrt{3-4x^2}}\,dx = P\arcsin(Qx) + c$ where $c$ is an arbitrary constant and $P$ and $Q$ are constants to be found. **(3 marks)**

**6** Evaluate:

**a** $\int_1^3 \frac{2}{1+x^2}dx$ **b** $\int_1^2 \frac{3}{\sqrt{1+4x^2}}dx$ **c** $\int_{-1}^2 \frac{1}{\sqrt{21-3x^2}}dx.$

E **7** Show that $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}}dx = \frac{\pi}{12}$ **(4 marks)**

E/P **8** $f(x) = \frac{2+3x}{1+3x^2}$

Show that $\int f(x)\,dx = A\arctan(\sqrt{3}x) + B\ln(1+3x^2) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(4 marks)**

**Hint** Start by splitting the fraction into two separate integrals.

E/P **9** $f(x) = \frac{2x-1}{\sqrt{2-x^2}}$

Find $\int f(x)\,dx$, giving your answer in the form $A\arcsin\left(\frac{x}{\sqrt{2}}\right) + B\sqrt{2-x^2} + c$ where $c$ is an arbitrary constant and $A$ and

$B$ are constants to be found. **(4 marks)**

E/P **10** $f(x) = \frac{8x-3}{4+x^2}$

Find $\int f(x)\,dx$, giving your answer in the form $A\ln(x^2+4) + B\arctan\left(\frac{x}{2}\right) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(4 marks)**

E/P **11** $f(x) = \frac{4x-1}{\sqrt{6-5x^2}}$

Show that $\int f(x)\,dx = P\sqrt{6-5x^2} + Q\arcsin\left(\sqrt{\frac{5}{6}}x\right) + c$ where $c$ is an arbitrary constant and $P$ and $Q$ are constants to be found. **(4 marks)**

E/P **12** $f(x) = \frac{x+5}{x^2+16}$

**a** Find $\int f(x)\,dx$, giving your answer in the form $A\ln(x^2+16) + B\arctan\left(\frac{x}{4}\right) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(4 marks)**

**b** Hence show that the mean value of $f(x)$ over the interval $[0, 4]$ is $\frac{1}{4}\left(\frac{1}{2}\ln 2 + \frac{5\pi}{16}\right)$ **(3 marks)**

**c** Hence write down the mean value of $-4f(x)$ over the interval $[0, 4]$. **(1 mark)**

(P) **13** Use the substitution $x = \frac{2}{3}\tan\theta$ to find $\int \frac{x^2}{9x^2+4}\,dx$.

(P) **14** By using the substitution $x = \frac{1}{2}\sin\theta$, show that $\int_0^{\frac{1}{4}} \frac{x^2}{\sqrt{1-4x^2}}\,dx = \frac{1}{192}(2\pi - 3\sqrt{3})$.

**Challenge**

Given that $x \geqslant 1$, use the substitution $x = \sec\theta$ to find:

**a** $\int \frac{1}{x\sqrt{x^2-1}}\,dx$ **b** $\int \frac{\sqrt{x^2-1}}{x}\,dx$

## 3.5 Integrating using partial fractions

In your A level course you used partial fractions to integrate some rational functions.

**Example 13**

Prove that

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + c$$

where $a$ is a real constant.

$$\int \frac{1}{a^2-x^2}\,dx = \int \frac{1}{(a+x)(a-x)}\,dx$$

Factorise the denominator in order to integrate using partial fractions. ← **Pure Year 2, Section 11.7**

$$\frac{1}{(a+x)(a-x)} = \frac{A}{a+x} + \frac{B}{a-x}$$

$$A(a-x) + B(a+x) = 1$$

$$x = -a \Rightarrow A = \frac{1}{2a}$$

$$x = a \Rightarrow B = \frac{1}{2a}$$

Split into partial fractions and calculate the values of $A$ and $B$ in terms of $a$.

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\int \frac{1}{a+x}\,dx + \frac{1}{2a}\int \frac{1}{a-x}\,dx$$

$$= \frac{1}{2a}\ln|a+x| - \frac{1}{2a}\ln|a-x| + c$$

Integrate using the reverse chain rule.

$$= \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + c$$

Simplify using the law of logarithms.

**Watch out** Be careful not to confuse this result with $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$

If the denominator of a partial fraction includes a quadratic factor of the form $(x^2 + c)$, $c > 0$, you cannot write it as a product of linear factors with real coefficients.

However, you can still write it in partial fractions, where the partial fraction corresponding to the quadratic factor has a linear numerator and quadratic denominator.

So $\dfrac{20}{(x+3)(x^2+1)} = \dfrac{A}{x+3} + \dfrac{Bx+C}{x^2+1}$

$20 = A(x^2+1) + (Bx+C)(x+3)$ — Find the values of $A$, $B$ and $C$ by multiplying both sides by $(x+3)(x^2+1)$.

Set $x = -3$: $20 = 10A \Rightarrow A = 2$ — Set $x = -3$ so that $(Bx+C)(x+3) = 0$

$A + B = 0 \Rightarrow B = -2$ — Equate coefficients of $x^2$ on each side.

$3B + C = 0 \Rightarrow C = 6$ — Equate coefficients of $x$ on each side.

So $\dfrac{20}{(x+3)(x^2+1)} = \dfrac{2}{x+3} + \dfrac{6-2x}{x^2+1}$

You can use the techniques from the previous section to integrate the second fraction on the right-hand side.

## Example 14

Show that $\displaystyle\int \frac{1+x}{x^3+9x}\,\mathrm{d}x = A\ln\left(\frac{x^2}{x^2+9}\right) + B\arctan\left(\frac{x}{3}\right) + c$, where $A$ and $B$ are constants to be found.

$\displaystyle\int \frac{1+x}{x^3+9x}\,dx = \int \frac{1+x}{x(x^2+9)}\,dx = \int\left(\frac{A}{x} + \frac{Bx+C}{x^2+9}\right)dx$ — Separate into partial fractions to facilitate integration.

$A(x^2+9) + Bx^2 + Cx = x + 1$

Equate $x^2$ terms: $A + B = 0$

Equate $x$ terms: $C = 1$

Equate constant terms: $9A = 1 \Rightarrow A = \frac{1}{9}, B = -\frac{1}{9}$ — Calculate the values of the coefficients by equating like terms.

$$\int \frac{1+x}{x^3+9x}\,dx = \int\left(\frac{1}{9x} + \frac{-\frac{1}{9}x+1}{x^2+9}\right)dx$$

$$= \int\left(\frac{1}{9x} - \frac{x-9}{9(x^2+9)}\right)dx$$

$$= \frac{1}{9}\int\frac{1}{x}\,dx - \frac{1}{9}\int\frac{x}{x^2+9}\,dx + \int\frac{1}{x^2+9}\,dx$$ — Separate into 3 fractions and integrate each separately.

$$= \frac{1}{9}\ln x - \frac{\frac{1}{9}\ln(x^2+9)}{2} + \frac{1}{3}\arctan\left(\frac{x}{3}\right) + c$$

$$= \frac{1}{9}\ln x - \frac{1}{18}\ln(x^2+9) + \frac{1}{3}\arctan\left(\frac{x}{3}\right) + c$$

$$= \frac{1}{18}(2\ln x - \ln(x^2+9)) + \frac{1}{3}\arctan\left(\frac{x}{3}\right) + c$$

Calculate the second integral using the reverse chain rule.
Calculate the third integral using $\displaystyle\int\frac{1}{a^2+x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$ with $a = 3$.

$$= \frac{1}{18}\ln\left(\frac{x^2}{x^2+9}\right) + \frac{1}{3}\arctan\left(\frac{x}{3}\right) + c$$ — Simplify using the laws of logarithms.

## Example 15

**a** Express $\dfrac{x^4 + x}{x^4 + 5x^2 + 6}$ as partial fractions.

**b** Hence find $\displaystyle\int \frac{x^4 + x}{x^4 + 5x^2 + 6}\,\mathrm{d}x$.

**a** $x^4 + 5x^2 + 6 = (x^2 + 2)(x^2 + 3)$

So $\dfrac{x^4 + x}{(x^2 + 2)(x^2 + 3)} \equiv A + \dfrac{Bx + C}{x^2 + 2} + \dfrac{Dx + E}{x^2 + 3}$

$x^4 + x \equiv A(x^2 + 2)(x^2 + 3) + (Bx + C)(x^2 + 3) + (Dx + E)(x^2 + 2)$

$A = 1$

$B + D = 0$

$3B + 2D = 1$

So $B = 1$ and $D = -1$

$5A + C + E = 0$

$6A + 3C + 2E = 0$

So $C = 4$ and $E = -9$

So $\dfrac{x^4 + x}{x^4 + 5x^2 + 6} \equiv 1 + \dfrac{x + 4}{x^2 + 2} - \dfrac{x + 9}{x^2 + 3}$

Start by factorising the denominator. Since $x^4 + 5x^2 + 6$ does not contain an $x^3$ term or an $x$ term, you can write it as $u^2 + 5u + 6$ where $u = x^2$.

The numerator and denominator both have degree 4, so this is an **improper fraction**. You will need a constant term, and terms with denominators $x^2 + 2$ and $x^2 + 3$.

Equate coefficients of $x^4$.

Equate coefficients of $x^3$ and $x$.

Equate coefficients of $x^2$ and constant terms, and use $A = 1$.

**Problem-solving**

There are other ways of determining the coefficients. You could find $A = 1$ by writing

$$\frac{x^4 + x}{x^4 + 5x^2 + 6} = \frac{(x^4 + 5x^2 + 6) - (5x^2 - x + 6)}{x^4 + 5x^2 + 6}$$

$$= 1 - \frac{5x^2 - x + 6}{x^4 + 5x^2 + 6}$$

Methods such as this are often quicker than using polynomial long division. You could also substitute $x = \mathrm{i}\sqrt{2}$ and $x = \mathrm{i}\sqrt{3}$ to eliminate terms.

**b** Using the partial fraction decomposition:

$$\int \frac{x^4 + x}{(x^2 + 2)(x^2 + 3)}\,dx = \int 1\,dx + \int \frac{x}{x^2 + 2}\,dx$$

$$+ \int \frac{4}{x^2 + 2}\,dx - \int \frac{x}{x^2 + 3}\,dx - \int \frac{9}{x^2 + 3}\,dx$$

Split the integral up into different parts that can be integrated using inverse trigonometric functions or the reverse chain rule.

$$= x + \frac{1}{2}\ln|x^2 + 2| + \frac{4}{\sqrt{2}}\arctan\left(\frac{x}{\sqrt{2}}\right)$$

$$- \frac{1}{2}\ln|x^2 + 3| - \frac{9}{\sqrt{3}}\arctan\left(\frac{x}{\sqrt{3}}\right) + c$$

Use $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$ and $\int \frac{1}{x^2 + a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$.

$$= x + \frac{1}{2}\ln\left|\frac{x^2 + 2}{x^2 + 3}\right| + 2\sqrt{2}\arctan\left(\frac{x}{\sqrt{2}}\right)$$

$$- 3\sqrt{3}\arctan\left(\frac{x}{\sqrt{3}}\right) + c$$

Simplify.

## Exercise 3E

**1** Express the following as partial fractions.

**a** $\frac{1}{(x^2 + 1)(x + 3)}$ **b** $\frac{1}{(x^2 + 2)(x - 1)}$ **c** $\frac{x - 4}{x(x^2 + 7)}$

E/P **2** Find $\int \frac{x^2 - 3x}{(x^2 + 6)(x + 2)}\,dx$, giving your answer in the form $A\ln|x + 2| + B\arctan\left(\frac{x}{\sqrt{6}}\right) + c$, where $A$ and $B$ are constants to be found, and $c$ is an arbitrary constant. **(4 marks)**

E/P **3** $f(x) = x^4 - x^3 - 4x^2 - 2x - 12$

**a** Given that $(x + 2)$ is a factor of $f(x)$, fully factorise $f(x)$. **(2 marks)**

**b** Hence find $\int \frac{x^3 - 20x^2 + 4x - 24}{x^4 - x^3 - 4x^2 - 2x - 12}\,dx$, giving your answer in the form $\ln\frac{\cdot x + 2^{\cdot A}}{\cdot x - 3^{\cdot B}} + D\arctan\left(\frac{x}{\sqrt{2}}\right) + c$, where $A$, $B$ and $D$ are constants to be found, and $c$ is an arbitary constant. **(5 marks)**

E/P **4** Find $\int \frac{2 - x}{x^3 + 4x}\,dx$, giving your answer in the form $A\ln\left(\frac{x^2}{x^2 + 4}\right) + B\arctan\left(\frac{x}{2}\right) + c$, where $A$ and $B$ are constants to be found, and $c$ is an arbitary constant. **(5 marks)**

E/P **5** Find $\int \frac{x^2 + 1}{4x^4 + 9x^2}\,dx$, giving your answer in the form $\frac{A}{x} + B\arctan\left(\frac{2x}{3}\right) + c$, where $A$ and $B$ are constants to be found, and $c$ is an arbitary constant. **(5 marks)**

E/P **6** Show that $\int \frac{x^3 + 9x^2 + x + 1}{x^4 - 1}\,dx = \ln \frac{|x-1|^A}{|x+1|^B} + D\arctan x + c$, where $A$, $B$ and $D$ are constants to be found, and $c$ is an arbitary constant. **(5 marks)**

E/P **7** $f(x) = x^3 - 4x^2 + 6x - 24$

**a** Given that $f(4) = 0$, fully factorise $f(x)$. **(2 marks)**

**b** Express $\frac{2x^2 - 3x + 24}{x^3 - 4x^2 + 6x - 24}$ in partial fractions. **(2 marks)**

**c** Use your answer to part **b** and an appropriate substitution to calculate $\int \frac{2x^2 - 3x + 24}{x^3 - 4x^2 + 6x - 24}\,dx$. **(4 marks)**

E/P **8** $f(x) = \frac{1}{(x-2)(2x-1)}$

**a** Calculate $\int f(x)\,dx$. **(3 marks)**

**b** Hence show that $\int_{\frac{1}{2}}^{2} \frac{1}{(x-2)(2x-1)}\,dx$ diverges. **(3 marks)**

E/P **9** **a** Express $\frac{x^4 + 5x^2 + 2x}{x^4 + 10x^2 + 24}$ as partial fractions. **(4 marks)**

**b** Hence find $\int \frac{x^4 + 5x^2 + 2x}{x^4 + 10x^2 + 24}\,dx$. **(5 marks)**

E/P **10** Use the method of partial fractions to find $\int \frac{x^2 + 4x + 10}{x^3 + 5x}\,dx$, $x > 0$. **(4 marks)**

E/P **11** Show that $\int_0^2 \frac{2}{(x+1)(x^2+1)}\,dx = \frac{1}{4}(\pi + 2\ln 2)$. **(4 marks)**

E/P **12** **a** Express $\frac{x^4 + 1}{x(x^2+2)^2}$ as partial fractions. **(4 marks)**

**b** Hence find $\int \frac{x^4 + 1}{x(x^2+2)^2}\,dx$. **(5 marks)**

**Challenge**

Find:

**a** $\int \frac{1}{x^2 - 8x + 8}\,dx$ **b** $\int \frac{1}{2x^2 + 4x + 11}\,dx$

**Problem-solving**

First complete the square in the denominator and then use an appropriate substitution.

## Mixed exercise 3

E **1 a** Using the substitution $u = e^x$, find $\int \frac{1}{e^x + e^{-x}} dx$. **(3 marks)**

**b** Hence show that $\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}} dx = \frac{\pi}{2}$ **(3 marks)**

E **2** Find the exact mean value of $f(x) = \frac{1 - \cos x}{\sin^2 x}$ over the interval $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. **(4 marks)**

E **3** Show that the exact mean value of $f(x) = x \sin 2x$ over the interval $\left[0, \frac{\pi}{2}\right]$ is $\frac{1}{2}$ **(4 marks)**

E **4 a** Find the derivative of $\arccos x^2$. **(3 marks)**

**b** Hence, or otherwise, calculate $\int \frac{3x}{\sqrt{16 - x^4}} dx$. **(1 mark)**

E/P **5** $f(x) = \arctan\left(\frac{2x + 3}{x - 1}\right)$

**a** Show that $f'(x) = -\frac{1}{x^2 + 2x + 2}$ **(4 marks)**

**b** Given that $-2 \leqslant x \leqslant 2$, show that $|f'(x)| \leqslant 1$ **(2 marks)**

P **6 a** Explain what it means for an integral to be improper.

**b** Identify two features of $\int_0^{\infty} \frac{1}{(x + 1)\sqrt{x}} dx$ which make it an improper integral.

**c** By differentiating $\arctan \sqrt{x}$, or otherwise, show that $\int_0^{\infty} \frac{1}{(x + 1)\sqrt{x}} dx$ is convergent and find its exact value.

E/P **7** $f(x) = \frac{1 + 5x}{\sqrt{1 - 5x^2}}$

Find $\int f(x)\, dx$, giving your answer in the form $A\sqrt{1 - 5x^2} + B \arcsin(\sqrt{5}\, x) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(4 marks)**

E **8 a** Show that $\int_0^t \frac{1}{x^2 + 1} dx = \arctan t$. **(2 marks)**

**b** Hence evaluate:

**i** $\int_0^{\infty} \frac{1}{x^2 + 1} dx$ **(2 marks)**

**ii** $\int_{-\infty}^{\infty} \frac{1}{x^2 + 1} dx$ **(2 marks)**

E/P **9** $f(x) = \dfrac{1 + 2x}{1 + 4x^2}$

**a** Find $\int f(x)\,dx$, giving your answer in the form $A\ln(1 + 4x^2) + B\arctan(2x) + c$ where $c$ is an arbitrary constant and $A$ and $B$ are constants to be found. **(4 marks)**

**b** Hence find the exact value of $\int_0^{0.5} f(x)\,dx$ **(3 marks)**

E **10** **a** Show that $\int \dfrac{1}{\sqrt{4 - 9x^2}}\,dx = P\arcsin Qx + c$ where $c$ is an arbitrary constant and $P$ and $Q$ are constants to be found. **(4 marks)**

**b** Hence show that $\int_0^{\frac{2}{3}} \dfrac{1}{\sqrt{4 - 9x^2}}\,dx = \dfrac{\pi}{6}$ **(3 marks)**

P **11** Use the substitution $x = \sin\theta$ to show that $\int_0^{\frac{1}{2}} \dfrac{x^4}{\sqrt{1 - x^2}}\,dx = \dfrac{(4\pi - 7\sqrt{3})}{64}$

E/P **12** $f(x) = \dfrac{x}{1 + x^4}$

**a** Use the substitution $u = x^2$ to calculate $\int_0^1 f(x)\,dx$. **(4 marks)**

**b** Hence show that $\int_0^{\infty} f(x)\,dx$ converges and state its value. **(3 marks)**

E/P **13** Show that $\int \dfrac{2x^3 - 2x^2 + 18x + 9}{x^4 + 9x^2}\,dx = A\ln|x| - \dfrac{B}{x} + D\arctan\left(\dfrac{x}{3}\right) + c$, where $A$, $B$ and $D$ are constants to be found. **(5 marks)**

E/P **14** $f(x) = \dfrac{x^2 - 3x + 14}{x^3 - 4x^2 + 2x - 8}$

**a** Express $f(x)$ in the form $\dfrac{P}{x - 4} + \dfrac{Q}{x^2 + 2}$, where $P$ and $Q$ are constants to be found. **(3 marks)**

**b** Find $\int f(x)\,dx$, giving your answer in the form $A\ln|x - 4| + B\arctan\left(\dfrac{x}{\sqrt{2}}\right) + c$ where $A$ and $B$ are constants to be found. **(4 marks)**

**c** Hence show that $\int_4^{\infty} f(x)\,dx$ diverges. **(2 marks)**

E/P **15** $f(x) = \dfrac{2}{x^3 + x}$

**a** Find $\int f(x)\,dx$. **(4 marks)**

**b** Hence show that the mean value of $f(x)$ over the interval $[1, 2]$ is $\ln\dfrac{8}{5}$ **(2 marks)**

**c** Hence, or otherwise, find the mean value, over the interval $[1, 2]$, of $2f(x) - \dfrac{6}{x}$ **(3 marks)**

## Challenge

A function is said to **attain its mean value** on the interval $[a, b]$ if there exists a value $c \in [a, b]$ such that $f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$.

**a** Show that the function $f(x) = x^3 - 2x + 4$ attains its mean value on the interval $[0, 2]$, and find the exact values of $c \in [0, 2]$ for which $f(c) = \frac{1}{2}\int_0^2 f(x)\,dx$.

**b** Give an example of a function which does not attain its mean value on the interval $[0, 2]$, fully justifying your answer.

## Summary of key points

**1** The integral $\int_a^b f(x)\,dx$ is **improper** if either:

- one or both of the limits is infinite
- $f(x)$ is undefined at $x = a$, $x = b$ or another point in the interval $[a, b]$.

**2** The **mean value** of the function $f(x)$ over the interval $[a, b]$, is given by

$$\frac{1}{b-a}\int_a^b f(x)\,dx$$

**3** If the function $f(x)$ has mean value $\bar{f}$ over the interval $[a, b]$, and $k$ is a real constant, then:

- $f(x) + k$ has mean value $\bar{f} + k$ over the interval $[a, b]$
- $kf(x)$ has mean value $k\bar{f}$ over the interval $[a, b]$
- $-f(x)$ has mean value $-\bar{f}$ over the interval $[a, b]$.

**4**

- $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

**5**

- $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c,\ a > 0,\ |x| < a$
- $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + c$

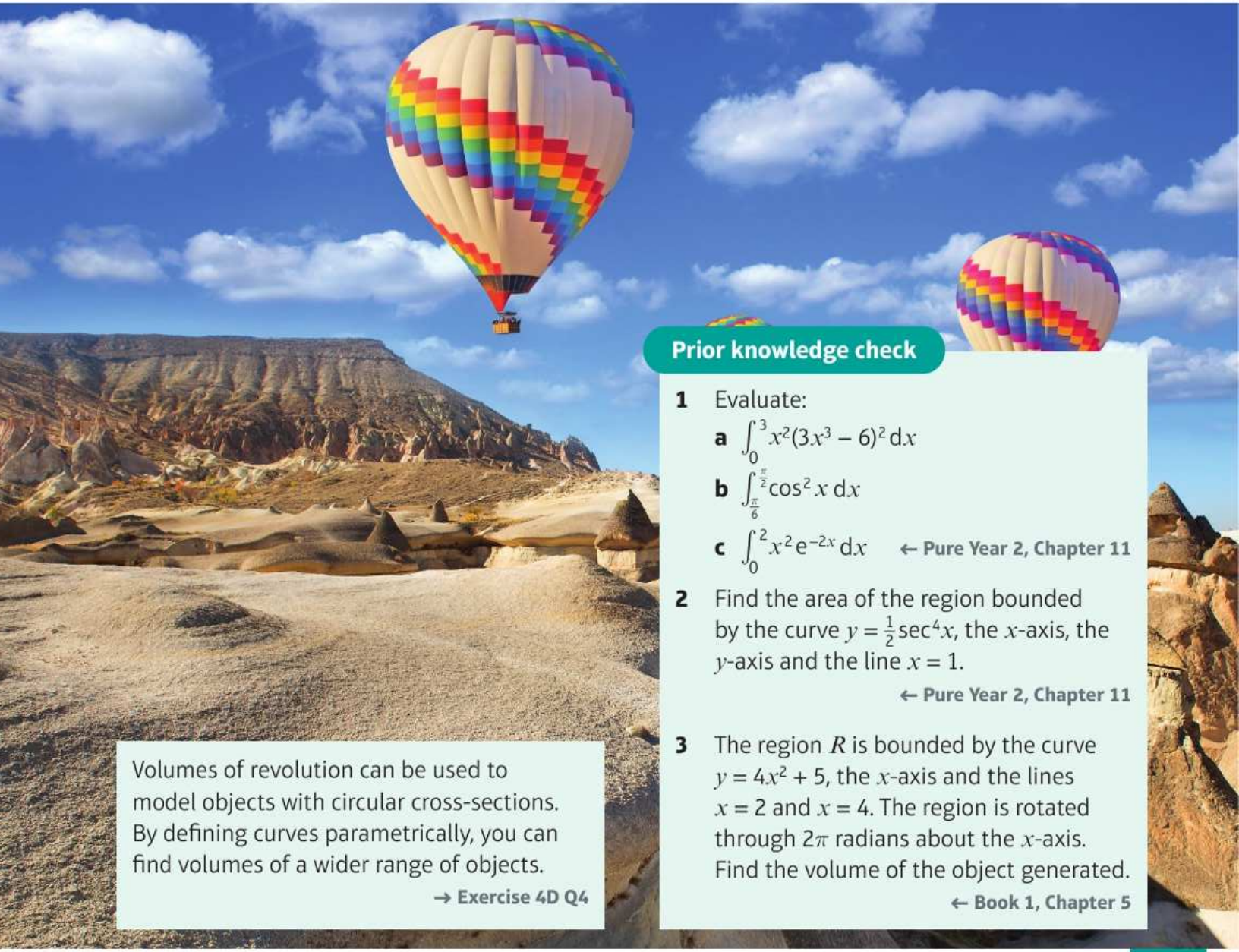
# Volumes of revolution

# 4

## Objectives

After completing this chapter you should be able to:

- Find volumes of revolution around the $x$-axis → pages 78–80
- Find volumes of revolution around the $y$-axis → pages 81–83
- Find volumes of revolution for curves defined parametrically → pages 83–87
- Model real-life applications of volumes of revolution → pages 87–89



Volumes of revolution can be used to model objects with circular cross-sections. By defining curves parametrically, you can find volumes of a wider range of objects.

→ Exercise 4D Q4

## Prior knowledge check

**1** Evaluate:

**a** $\int_0^3 x^2(3x^3 - 6)^2 \, dx$

**b** $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 x \, dx$

**c** $\int_0^2 x^2 e^{-2x} \, dx$ ← Pure Year 2, Chapter 11

**2** Find the area of the region bounded by the curve $y = \frac{1}{2}\sec^4 x$, the $x$-axis, the $y$-axis and the line $x = 1$.

← Pure Year 2, Chapter 11

**3** The region $R$ is bounded by the curve $y = 4x^2 + 5$, the $x$-axis and the lines $x = 2$ and $x = 4$. The region is rotated through $2\pi$ radians about the $x$-axis. Find the volume of the object generated.

← Book 1, Chapter 5